

A self-adaptive frequency selection common spatial pattern and least squares twin support vector machine for motor imagery electroencephalography recognition

Duan Li^{a,b,*}, Hongxin Zhang^a, Muhammad Saad Khan^a, Fang Mi^a

^a School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China

^b School of Physics and Electronic Information Engineering, Henan Polytechnic University, Jiaozuo, 454000, China



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ABSTRACT

Motor imagery brain-computer interface (BCI) systems require accurate and fast recognition of brain activity patterns for reliable communication and interaction. Achieving this accuracy is still a challenge because of the low signal-to-noise ratio in electroencephalography signals and high variability of sensorimotor rhythms. To address this need, we proposed a novel scheme that combined a frequency band selection common spatial pattern algorithm and a particle swarm optimization least squares twin support vector machine classifier for recognition of motor imagery patterns. We used self-adaptive artifact removal and the common spatial pattern method to obtain the most distinguishable features. To improve the classification results, we investigated linear, polynomial, sigmoid, wavelet and gaussian kernel functions for the nonlinear least squares twin support vector machine classifier. Particle swarm optimization, chaotic particle swarm optimization, a genetic algorithm and a quantum genetic algorithm were compared and used to tune the hyper-parameters for the classifier. To evaluate our proposed method, we used BCI Competition IV data sets 2A and 2B. Experimental results showed that for our method, the average recognition accuracy of data set 2A is increased by 6.10%, 6.71%, 3.87%, 4.01%, 2.55% and 4.86% compared with the results obtained by regularization projection twin support vector machine, twin support vector machine, support vector machine, linear discriminant analysis, back propagation and probabilistic neural network, respectively. Using data set 2B, the average recognition accuracy achieved by our method was greater by 4.73%, 5.46%, 4.45%, 4.10%, 8.62% and 4.27%, respectively. The standard deviation of the accuracy values of the two data sets decreased. Furthermore, compared with the traditional support vector machines, our proposed method achieved a faster central processing unit running time for training classifiers.

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1. Introduction

Brain-computer interface (BCI) research is an emerging field that explores the use of electrical brain activity to control devices external to the body. There are a wide variety of applications for BCI in diverse areas, ranging from control of neuroprosthetics for rehabilitation to operation of semi-autonomous cars, entertainment devices, and military equipment [1,2]. Motor imagery BCI (MI-BCI) systems are designed to detect and decode a user's imagined images of a motor task, such as the imagined hand movement needed to roll a wheelchair forward or to move a cursor left or right

on a computer screen. The MI-BCI then creates output that implement the desired action. There are many invasive and non-invasive approaches to acquiring and recording brain signals. Among the non-invasive methods, electroencephalography (EEG) has proven to be the best recording technique because of its excellent temporal resolution, usability, low set-up costs, and widespread availability [3–7].

A subject's imagined movements of different body parts can cause power decrease in the sensorimotor rhythms of the EEG at corresponding “active” cortex areas, i.e., μ (8–13 Hz) and β (18–30 Hz) rhythms, called event-related desynchronization (ERD). Meanwhile, a power increase in sensorimotor rhythms, called event-related synchronization (ERS), might be observed at other “idling” areas during motor imagery [8,9]. Though these frequency bands are useful for detecting MI tasks, the ERD/ERS patterns exhibit variability across subjects. In fact, variability has

* Corresponding author at: No.10, Xitu Street, Haidian District, Beijing 100876, China.

E-mail addresses: lijuan@hpu.edu.cn, 15650713572@163.com (D. Li).

been detected even across trials for the same subject. Furthermore, EEG signals may be contaminated with various artifacts, such as electromyogram (EMG) and electro-oculogram (EOG) signals. Therefore, it is necessary to perform EEG signal preprocessing and feature extraction in order to represent input signals in a reduced feature space and improve MI pattern recognition. A variety of feature extraction methods have been proposed [10–14]. Among them, common spatial pattern (CSP) feature extraction is a supervised algorithm for learning spatial filters, and it has achieved a high performance in the multi-channel EEG signal filter in recent years [15–17]. The ERD/ERS patterns in the corresponding frequency bands are distinct, but it is variable for different subjects. Subject-specific frequency band selection is necessary for feature extraction which can improve the inter-subject robustness of MI-based BCI [18].

The core of an MI-BCI system is its discrimination algorithm. Many useful discriminant methods have been proposed to improve the robustness and accuracy of BCIs, such as the locality preserving projection method based on a self-regression model [19], the linear discriminant algorithm [20], the probabilistic methods [21,22], and various neural networks [23–25]. Support vector machines (SVMs) have also been used frequently for classification of EEG patterns, including an SVM combined with intelligent optimization algorithms that were certified to give high quality results in BCI [14,26,27].

The traditional SVM involves the solution of a single quadratic programming problem (QPP). This approach can be time-consuming for data sets that have a large number of features. Also, an SVM involves obtaining the predicted label using a single maximum-margin hyperplane. The twin SVM (TWSVM), first proposed by Jayadeva et al. [28], is based on the idea that a better prediction can be obtained using two non-parallel hyperplanes. TWSVMs are 4 times faster than conventional SVMs, and they have a solid theoretical foundation with promising generalization [29–31]. Recently, TWSVMs have been used successfully in the bioinformatics field [32–34]. Many scholars have contributed to the study of TWSVMs, achieving notable results [35–41].

In real-world applications, EEG signals are non-stationary and are often characterized by their shifting nature [17]. The TWSVM classifier is suitable for shifting data because it aims to find two non-parallel hyperplanes to classify the testing data. In contrast, the least squares twin SVM (LSTSVM) considers equality constraints rather than inequality constraints, making it simple to program complex quadratic problems, an approach that also allows for easier real-time hardware implementation. If the kernel parameters and penalty parameters are appropriate, the LSTSVM classifier will achieve better generalization ability, as compared with the TWSVM classifier [42–46]. For our research, we designed a more robust, high-efficiency motor imagery recognition system by introducing an LSTSVM classifier for motor imagery pattern recognition. To the best of our knowledge, this approach is novel to this field.

In our study, we used adaptive EOG artifact removal and a self-adaptive frequency band selection CSP algorithm for EEG preprocessing and feature extraction. Because the parameters of the classifier can affect the classification results significantly, we tuned the parameters of our proposed classifier by comparing our method with a genetic algorithm (GA) and particle swarm optimization (PSO) and also with their improved methods, a quantum genetic algorithm (QGA) and chaotic particle swarm optimization (CPSO). Furthermore, we compared traditional SVMs and other machine learning methods with our proposed method in terms of classification results and CPU running time.

The remainder of this paper is organized as follows. Section 2 described the related work and methods for this study. In section 3, we provide details of our novel self-adaptive preprocessing, feature extraction and bio-inspired optimization techniques based on

an LSTSVM scheme for MI-BCI. Our experimental results and discussion are provided in Section 4 and Section 5, respectively. Section 6 concludes the paper.

2. Methods

2.1. Least squares twin SVM

Consider a binary classification problem of classifying m_1 data points belonging to class +1 and m_2 data points belonging to class –1 in the n -dimensional real space $\mathbb{R}^n$. Let matrix $\mathbf{X}_1$ in $\mathbb{R}^{m_1 \times n}$ represent the data points of class +1 and matrix $\mathbf{X}_2$ in $\mathbb{R}^{m_2 \times n}$ represent the data points of class –1. Linear LSTSVM seeks two non-parallel hyperplanes in $\mathbb{R}^n$, as shown in equation (1).

$$\begin{cases} \chi \mathbf{w}_1 + b_1 = 0 \\ \chi \mathbf{w}_2 + b_2 = 0 \end{cases} \quad (1)$$

Instead of inequality constraints, LSTSVM determines two non-parallel hyper-planes by solving the following pair of linear equations (2) and (3).

$$\begin{cases} \min_{\mathbf{w}_1, b_1, \xi} \frac{1}{2} \|\mathbf{X}_1 \mathbf{w}_1 + \mathbf{e}_1 b_1\|^2 + \frac{c_1}{2} \xi^T \xi \\ \text{s.t. } -(\mathbf{X}_2 \mathbf{w}_1 + \mathbf{e}_2 b_1) + \xi = \mathbf{e}_2 \end{cases} \quad (2)$$

and

$$\begin{cases} \min_{\mathbf{w}_2, b_2, \eta} \frac{1}{2} \|\mathbf{X}_2 \mathbf{w}_2 + \mathbf{e}_2 b_2\|^2 + \frac{c_2}{2} \eta^T \eta \\ \text{s.t. } (\mathbf{X}_1 \mathbf{w}_2 + \mathbf{e}_1 b_2) + \eta = \mathbf{e}_1 \end{cases} \quad (3)$$

where $\mathbf{e}_1 \in \mathbb{R}^{m_1 \times 1}$ and $\mathbf{e}_2 \in \mathbb{R}^{m_2 \times 1}$ represent the vector of 1's. ξ , η are slack variables and c_1, c_2 are positive penalty parameters. The solution of equations (2) and (3) determines hyper-plane parameters as shown in formulas (4) and (5).

$$\begin{bmatrix} \mathbf{w}_1 \\ b_1 \end{bmatrix} = -\left(\mathbf{B}^T \mathbf{B} + \frac{1}{c_1} \mathbf{A}^T \mathbf{A}\right)^{-1} \mathbf{B}^T \mathbf{e}_2 \quad (4)$$

and

$$\begin{bmatrix} \mathbf{w}_2 \\ b_2 \end{bmatrix} = \left(\mathbf{A}^T \mathbf{A} + \frac{1}{c_2} \mathbf{B}^T \mathbf{B}\right)^{-1} \mathbf{A}^T \mathbf{e}_1 \quad (5)$$

Where $\mathbf{A} = [\mathbf{X}_1 \ \mathbf{e}_1]$ and $\mathbf{B} = [\mathbf{X}_2 \ \mathbf{e}_2]$. Thus the linear LSTSVM completely solves the classification problem with just two matrix inverses.

Once the weights and biases of the two non-parallel separating hyperplanes as given in Equation (1) are obtained from equation (4) and (5), LSTSVM classifier assigns the class according to the distance of a given data point from its corresponding hyper-plane. The point is classified into a class which lies nearest to it. LSTSVM predicts the class according to the following decision function (6).

$$f(\chi) = \arg \min_{i=+1, -1} \frac{|\chi \mathbf{w}_i + b_i|}{\|\mathbf{w}_i\|} \quad (6)$$

where $|\cdot|$ denotes the absolute value.

LSTSVM can also be extended to handle nonlinear kernels by considering two non-parallel kernel generated surfaces as shown in equation (7).

$$\begin{cases} K(\chi, \mathbf{X}) \mathbf{u}_1 + \gamma_1 = 0 \\ K(\chi, \mathbf{X}) \mathbf{u}_2 + \gamma_2 = 0 \end{cases} \quad (7)$$

where $\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}$ and K is any arbitrary kernel.

The primal QPPs of nonlinear LTSVM can be modified in the same way with the 2-norm of slack variables and inequality constraints replaced by equality constraints as shown in eqs. (8) and (9).

$$\begin{cases} \min_{\mathbf{u}_1, \gamma_1, \xi} & \frac{1}{2} \|K(\mathbf{X}_1, \mathbf{X})\mathbf{u}_1 + \mathbf{e}_1\gamma_1\|^2 + \frac{c_1}{2} \xi^T \xi \\ \text{s.t.} & -(K(\mathbf{X}_2, \mathbf{X})\mathbf{u}_1 + \mathbf{e}_2\gamma_1) + \xi = \mathbf{e}_2 \end{cases} \quad (8)$$

and

$$\begin{cases} \min_{\mathbf{u}_2, \gamma_2, \eta} & \frac{1}{2} \|K(\mathbf{X}_2, \mathbf{X})\mathbf{u}_2 + \mathbf{e}_2\gamma_2\|^2 + \frac{c_2}{2} \eta^T \eta \\ \text{s.t.} & (K(\mathbf{X}_1, \mathbf{X})\mathbf{u}_2 + \mathbf{e}_1\gamma_2) + \eta = \mathbf{e}_1 \end{cases} \quad (9)$$

By substituting the constraints into the objective function, these QPPs become the following formulas (10) and (11).

$$\min_{\mathbf{u}_1, \gamma_1} \left[\frac{1}{2} \|K(\mathbf{X}_1, \mathbf{X})\mathbf{u}_1 + \mathbf{e}_1\gamma_1\|^2 + \frac{c_1}{2} \|K(\mathbf{X}_2, \mathbf{X})\mathbf{u}_1 + \mathbf{e}_2\gamma_1 + \mathbf{e}_2\|^2 \right] \quad (10)$$

$$\begin{aligned} \min_{\mathbf{u}_2, \gamma_2} & \left[\frac{1}{2} \|K(\mathbf{X}_2, \mathbf{X})\mathbf{u}_2 + \mathbf{e}_2\gamma_2\|^2 \right. \\ & \left. + \frac{c_2}{2} \| -K(\mathbf{X}_1, \mathbf{X})\mathbf{u}_2 - \mathbf{e}_1\gamma_2 + \mathbf{e}_1 \|^2 \right] \end{aligned} \quad (11)$$

The solution of QPPs (10) and (11) can be derived to be equations (12) and (13).

$$\begin{bmatrix} \mathbf{u}_1 \\ \gamma_1 \end{bmatrix} = - \left(\mathbf{H}^T \mathbf{H} + \frac{1}{c_1} \mathbf{G}^T \mathbf{G} \right)^{-1} \mathbf{H}^T \mathbf{e}_2 \quad (12)$$

and

$$\begin{bmatrix} \mathbf{u}_2 \\ \gamma_2 \end{bmatrix} = \left(\mathbf{G}^T \mathbf{G} + \frac{1}{c_2} \mathbf{H}^T \mathbf{H} \right)^{-1} \mathbf{G}^T \mathbf{e}_1 \quad (13)$$

Where $\mathbf{G} = [K(\mathbf{X}_1, \mathbf{X}) \mathbf{e}_1]$ and $\mathbf{H} = [K(\mathbf{X}_2, \mathbf{X}) \mathbf{e}_2]$.

It can be noted that the solution of nonlinear LTSVM requires inversion of matrix of size $(n+1) \times (n+1)$ twice. To reduce the computation complexity, using Sherman-Morrison-Woodbury (SMW) formula, the nonlinear LTSVM can be solved by three inverse of smaller dimension than $(n+1) \times (n+1)$ [47]. The solution of nonlinear LTSVM for two cases. For the case of $m_1 < m_2$, the (12) and (13) can be rewritten as equations (14) and (15).

$$\begin{bmatrix} \mathbf{u}_1 \\ \gamma_1 \end{bmatrix} = -(\mathbf{Y} - \mathbf{Y}\mathbf{G}^T(c_1\mathbf{I} + \mathbf{G}\mathbf{Y}\mathbf{G}^T)^{-1}\mathbf{G}\mathbf{Y})\mathbf{H}^T\mathbf{e}_2 \quad (14)$$

$$\begin{bmatrix} \mathbf{u}_2 \\ \gamma_2 \end{bmatrix} = c_2(\mathbf{Y} - \mathbf{Y}\mathbf{G}^T(\frac{1}{c_2} + \mathbf{G}\mathbf{Y}\mathbf{G}^T)^{-1}\mathbf{G}\mathbf{Y})\mathbf{G}^T\mathbf{e}_1 \quad (15)$$

where $\mathbf{Y} = (\mathbf{H}^T \mathbf{H})^{-1}$. To solve the problems of possible ill-conditioning of $\mathbf{H}^T \mathbf{H}$, we introduced a regularization term $\varepsilon \mathbf{I}, \varepsilon > 0$ to $\mathbf{Y}$. So $\mathbf{Y} = \frac{1}{\varepsilon} (\mathbf{I} - \mathbf{H}^T (\varepsilon \mathbf{I} + \mathbf{H}\mathbf{H}^T)^{-1} \mathbf{H})$.

In an exactly similar way, for the case of $m_2 < m_1$, we rewrote (12) and (13) as equations (16) and (17).

$$\begin{bmatrix} \mathbf{u}_1 \\ \gamma_1 \end{bmatrix} = -c_1(\mathbf{Z} - \mathbf{Z}\mathbf{H}^T(\frac{1}{c_1} + \mathbf{H}\mathbf{Z}\mathbf{H}^T)^{-1}\mathbf{H}\mathbf{Z})\mathbf{H}^T\mathbf{e}_2 \quad (16)$$

$$\begin{bmatrix} \mathbf{u}_2 \\ \gamma_2 \end{bmatrix} = (\mathbf{Z} - \mathbf{Z}\mathbf{H}^T(c_2\mathbf{I} + \mathbf{H}\mathbf{Z}\mathbf{H}^T)^{-1}\mathbf{H}\mathbf{Z})\mathbf{G}^T\mathbf{e}_1 \quad (17)$$

where $\mathbf{Z} = \frac{1}{\varepsilon} (\mathbf{I} - \mathbf{G}^T (\varepsilon \mathbf{I} + \mathbf{G}\mathbf{G}^T)^{-1} \mathbf{G})$.

In this way, a non-linear binary LTSVM classifier generates kernel surfaces for each class in a higher dimensional space. For a test data point, its perpendicular distance is calculated from each kernel generated surface and the classifier predicts the class from which the distance of the point is least. The final decision function of a non-linear classifier as shown in equation (18).

$$f(\mathbf{x}) = \arg \min_{i=1,2} \frac{|K(\mathbf{x}, \mathbf{X})\boldsymbol{\mu}_i + \gamma_i|}{\|\boldsymbol{\mu}_i\|} \quad (18)$$

2.2. Kernel functions for non-linear LTSVM

As mentioned above, we can project the samples to nonlinear high dimensional space using kernel functions, an approach that can greatly improve the performance of the classifier. Similar to traditional SVMs [48,49], the kernel function selection will affect the classification performance and generalization ability of the LTSVM. To obtain the best experimental results, we investigated linear, polynomial, sigmoid, wavelet and gaussian kernel functions. The kernel functions are defined as formulas (19–23), respectively.

$$K(\mathbf{x}, \mathbf{z}) = \mathbf{x} \cdot \mathbf{z}^T \quad (19)$$

$$K(\mathbf{x}, \mathbf{z}) = (\mathbf{x} \cdot \mathbf{z}^T + c)^d \quad (20)$$

$$K(\mathbf{x}, \mathbf{z}) = \tanh(g \cdot \mathbf{x} \cdot \mathbf{z}^T + c) \quad (21)$$

$$K(\mathbf{x}, \mathbf{z}) = (d - \sum_{i=1}^d (\frac{x_i - z_i}{a_i})^2) \exp(-\frac{1}{2} \sum_{i=1}^d (\frac{x_i - z_i}{a_i})^2) \quad (22)$$

$$K(\mathbf{x}, \mathbf{z}) = e^{-\|\mathbf{x} - \mathbf{z}\|^2 / 2\sigma^2} \quad (23)$$

where $\mathbf{x}, \mathbf{z} \in \mathbb{R}^d$.

2.3. Optimization algorithms

2.3.1. Particle swarm optimization and chaotic particle swarm optimization

PSO is a parallel evolutionary computation technique [50], firstly introduced by Kennedy and Eberhart in 1995. A swarm consists of n particles; each particle has m -dimensions, a position vector $\mathbf{X}_i = (x_{i1}, x_{i2}, \dots, x_{im})$, and a velocity vector $\mathbf{V}_i = (v_{i1}, v_{i2}, \dots, v_{im})$, where $i = 1, 2, \dots, n$. During each generation, each particle is accelerated toward its previous best position and the global best position. Let $\mathbf{p}_i = (p_{i1}, p_{i2}, \dots, p_{im})$ represent the individual optimum value, and $\mathbf{p}_g = (p_{g1}, p_{g2}, \dots, p_{gm})$ represent the global optimum value. The new velocity value is then used to calculate the next position of the particle in the search space. Updating the velocity and particle position can be obtained by using the formulas (24) and (25).

$$v_{id}^{k+1} = w \times v_{id}^k + c_1 \times r_1 \times (p_{id} - x_{id}^k) + c_2 \times r_2 \times (p_{gd} - x_{id}^k) \quad (24)$$

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1} \quad (25)$$

Where $i = 1, 2, \dots, n$ and $d = 1, 2, \dots, m$. r_1 and r_2 are random variables in the range $[0, 1]$, c_1 and c_2 are acceleration constants. These parameters are considered as scaling factors that determine the best position of the particle and the global best position.

To avoid premature convergence to a local optimal solution, many studies combine PSO and chaos theory to solve various complex optimization problems [51,52]. Compared with the standard PSO, in the initialization phase of CPSO, chaotic initialization is used to select better initial positions for the particles instead of random selection, thereby increasing the population diversity and ergodicity in the search. In the optimal position search process, chaotic queues are generated based on the global optimal position searched

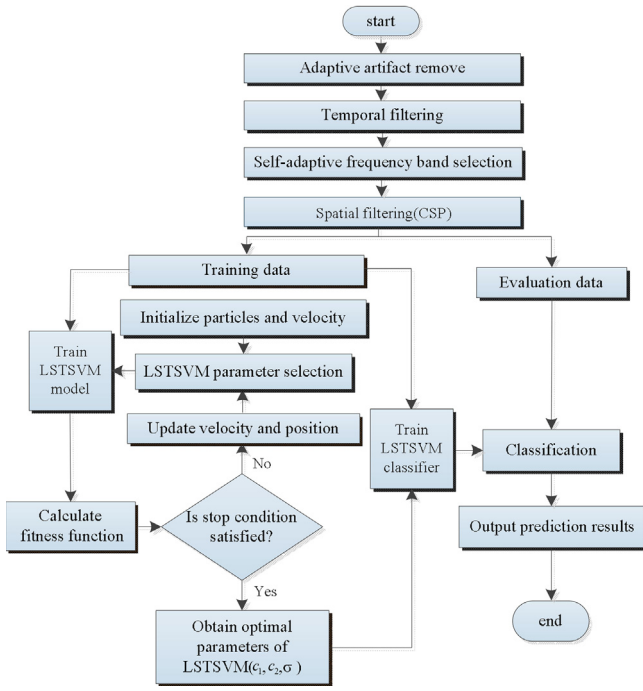


Fig. 1. Algorithm of PSO LSTSVM based MI recognition.

by all particles until a given point in time, then the particle position in the population is replaced by the best position in the chaotic queue.

2.3.2. Genetic algorithms and quantum genetic algorithms

The GA is a biology-inspired search technique that has been used successfully to solve complex problems in numerous engineering domains. The parameter search process for a classifier is shown as follows:

Step1: Encode parameters of the classifier and initialization of population.

Step2: Calculate the fitness function of each chromosome.

Step3: Apply GA operators: selection, crossover and mutation.

Step4: Generate a new population for the next generation. Offspring replaces the old population and forms a new population in the next generation.

Step5: Obtain the parameters of classifier model. If the stop criteria are satisfied, the best chromosomes are presented as a solution, otherwise return to Step 2 and repeat.

The quantum GA is based on the basic framework of the classical GA. A typical iteration of the QGA consists of the cyclic application of selection, variation operators (quantum crossover, quantum mutation, and quantum interference), quantum measurement, evaluation and replacement. For more detailed information about quantum GAs, please refer to the work presented in [53,54].

3. Application to motor imagery BCI recognition

3.1. Implementation scheme

In our research, for EEG signal preprocessing and feature extraction, we adopted EOG artifact removal and a self-adaptive frequency band selection CSP filter for motor imagery (MI), respectively. The different optimal algorithm-based LSTSVM classifiers were applied to classify the MI patterns using the BCI competition IV 2A and 2B data sets. Fig. 1 exhibits the flow of our proposed PSO LSTSVM classifier algorithm.

3.2. Data analysis

Our data analysis included examination the following data sets, as described below.

3.2.1. BCI Competition IV data set 2A

BCI Competition IV data set 2A included EEG signals from 9 subjects (A01–A09), and each subject provided 2 recording sessions. The data consisted of 25 channels, including 22 EEG channels, and 3 monopolar EOG channels. The signals were sampled with 250 Hz and bandpass-filtered between 0.5 Hz and 100 Hz. The EOG channels are provided for the subsequent application of artifact processing methods. These signals included 4 classes of motor imagery: left hand (class1), right hand (class 2), both feet (class 3), and tongue (class 4). Each session consists of six runs separated by short breaks and each run comprised of 48 trials (12 for each class). The total numbers of 288 trials are in each session. Only the left hand and right hand motor imagery patterns were considered in this paper. Section I data were used for training, and Section II data were used for evaluating the classifier.

3.2.2. BCI Competition IV data set 2B

BCI Competition IV Data set 2B comprised EEG measurements from 9 subjects (B01–B09), with 5 sessions recorded from each subject. Each session consisted of EEG data recorded from 3 bipolar recordings (C3, Cz, C4) and 3 monopolar EOG channels, which were acquired with a sampling frequency of 250 Hz. These signals included 2 classes of motor imagery: left hand and right hand. The motor imagery data from Sessions I, II and III were used to train our classifier, and the data from Sessions IV and V were used to evaluate the performance of the test.

3.3. Data preprocessing and feature extraction

3.3.1. Self-adaptive artifact removal

EEG technique is sensitive to noise generated by other physiological sources. Consequently, it is necessary to remove unwanted artifacts. For this research, we used an adaptive EOG artifact removal technique. Taking data set 2A as an example, the 3 EOG channels were used for EEG artifact removal. Suppose that the 3 EOG channels are denoted by $\mathbf{o}_1$, $\mathbf{o}_2$ and $\mathbf{o}_3$. The model is shown in equation (26):

$$\mathbf{a}(t) = \mathbf{j}(t) + \{\mathbf{o}_1(t), \mathbf{o}_2(t), \mathbf{o}_3(t)\}\mathbf{b} \quad (26)$$

where $\mathbf{a}(t)$ is the recorded value of a channel, $\mathbf{j}(t)$ is the source signal without artifact contamination, and $\mathbf{b} \in R^{3 \times 1}$ represents the weight of the EOG artifacts at this EEG channel. Then, the multi-channel EEG signals can be written as shown in equation (27).

$$\mathbf{A}_{T \times M} = \mathbf{J}_{T \times M} + \mathbf{O}_{T \times N}\mathbf{B}_{N \times M} \quad (27)$$

The indices indicate the size of each matrix. The signals $\mathbf{A}$ and $\mathbf{J}$ have T time points and M channels. The noise $\mathbf{O}$ has N components, and $\mathbf{B}$ denotes the weights from each EOG component to each EEG channel. Therefore, formula (27) can be rewritten in the following manner:

$$\mathbf{J} = \mathbf{A} - \mathbf{O}\mathbf{B} \quad (28)$$

then

$$\mathbf{O}^T \mathbf{J} = \mathbf{O}^T \mathbf{A} - \mathbf{O}^T \mathbf{O}\mathbf{B} \quad (29)$$

Since the signal $\mathbf{J}$ and the noise $\mathbf{O}$ are independent, their cross-covariance equals zero. Thus, the weights $\mathbf{B}$ can be represented in equation (30).

$$\mathbf{B} = \mathbf{D}_{NN}^{-1} \mathbf{D}_{NY} \quad (30)$$

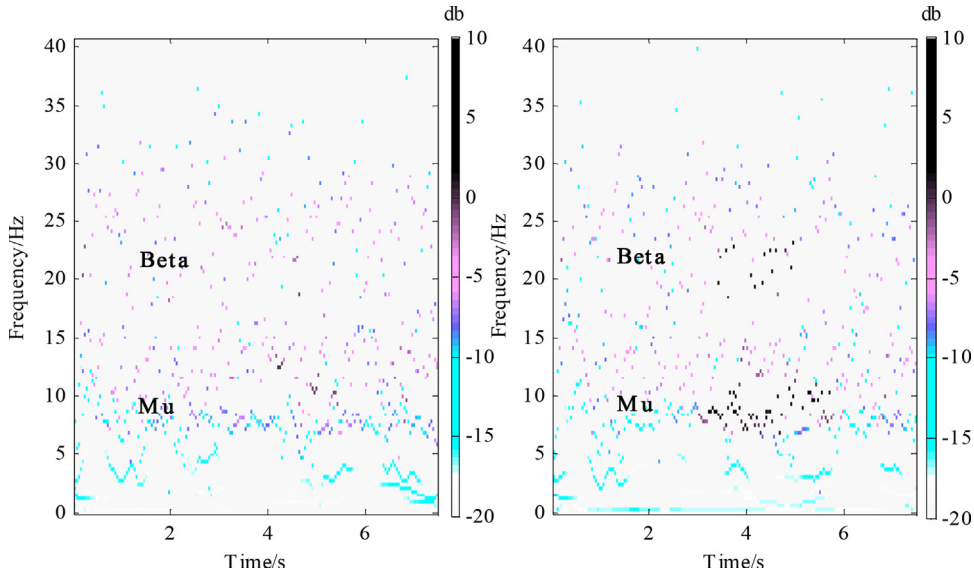


Fig. 2. ERSP for right hand MI task of subject A08.

(a) ERSP for the left hemisphere(c3) (b) ERSP for the right hemisphere(c4).

where $\mathbf{D}_{NN}$ is the auto-covariance matrix of the EOG channels, and $\mathbf{D}_{NY}$ is the cross-covariance between the EEG channel and EOG channel.

3.3.2. Spectral perturbation analysis and filtering

Based on the neurophysiological mechanisms, the *mu* and *beta* rhythms in an EEG can be observed in the central region of the brain when subjects plan and execute hand or finger movements. Each trial of the two data sets began with a fixation cross. Some seconds later, a visual cue was presented, and the subject was asked to imagine the corresponding task over a period of 4 s. The duration of each subject's focus on the fixation cross was 2 s in data set 2A and 3 s in data set 2B. In the following experiments, the time segment for each MI task ranged from 2.6s to 6.8 s for data set 2A, and from 3 s to 7.6 s for data set 2B.

Fig. 2 illustrates the event-related spectral perturbation (ERSP) of the right hand MI task of subject A08 in data set 2A. The time-amplitude changes for the right hand MI of A08 from electrode C3 and C4 are shown in Fig. 3. Time scope was 1–7s, including the whole MI task process. The frequency band covered 0–40 Hz. The ERSP illustrates obvious contralateral decrease (ERD) and ipsilateral increase (ERS), especially in the *mu* (8–12 Hz) bands. As shown in Fig. 2 and Fig. 3, the ERS and ERD within the *mu* rhythms were more prominent than those within the *beta* rhythms for subject A08.

In the above analysis, the (8–12) Hz (*mu* band) and (14–30) Hz (*beta* band) showed a distinct frequency response related to MI-associated phenomena of event-related synchronization and desynchronization for subject A08. The frequency bands varied across subjects because each subject had different psychological, neuro-physiological, and brain neuroanatomic characteristics. To adapt such variations for MI-BCIs based on sensorimotor-rhythms. Several 5th order Butterworth filters were employed to obtain 10 frequency bands: 8–16, 8–24, 8–32, 8–40, 16–24, 16–32, 16–40, 24–32, 24–40 and 32–40 Hz. The frequency band could be selected self-adaptively for different subjects. The optimal frequency band for a specific subject was obtained by using 10-fold cross-validation (10-CV) classification accuracy on the training data sets.

3.3.3. Common spatial pattern feature extraction

In this part of our work, we used common spatial pattern feature extraction (CSP) to filter the EEG signals of the various frequency bands that had been preprocessed in the previous stage. The CSP

approach aims to find a spatial filter to maximize the variance for another class, while at the same time achieving the purpose of classification. Assume the original EEG signal are represented as a matrix $E \in R^{N \times T}$, where N represents the number of channels, and T denotes the sample points of each channel. The spatial covariance of the EEG data can be calculated by equation (31).

$$\mathbf{C} = \mathbf{E}\mathbf{E}^T / \text{tr}(\mathbf{E}\mathbf{E}^T) \quad (31)$$

In order to separate the two classes, two types of data are required on average, respectively. After getting the average covariance $\mathbf{C}_1$ and $\mathbf{C}_2$, a composite spatial covariance is obtained as shown in equation (32).

$$\bar{\mathbf{C}} = \bar{\mathbf{C}}_1 + \bar{\mathbf{C}}_2 \quad (32)$$

Decompose $\bar{\mathbf{C}}$ into:

$$\bar{\mathbf{C}} = \mathbf{V}_c \mathbf{L} \mathbf{V}_c^T \quad (33)$$

where $\mathbf{V}_c$ is a decomposed feature vector, and $\mathbf{L}$ is a diagonal matrix composed of eigenvalues. Arrange the eigenvalues in descending order, and then transform them by whitening as shown in equation (34).

$$\mathbf{Q} = \sqrt{\mathbf{L}^{-1}} \mathbf{V}_c^T \quad (34)$$

Then, the covariance matrices can be transformed into the following expressions in equation (35).

$$\begin{cases} \mathbf{Q}\bar{\mathbf{C}}_i\mathbf{Q}^T = \mathbf{S}_i = \mathbf{V}\mathbf{L}_i\mathbf{V}^T \\ \mathbf{L}_1 + \mathbf{L}_2 = \mathbf{I} \end{cases} \quad (35)$$

The projection matrix is achieved by the following equation (36).

$$\mathbf{W} = (\mathbf{V}^T \mathbf{Q})^T \quad (36)$$

After whitening, the EEG signals can be projected on the first m and last m columns of $\mathbf{B}$. Thus, the EEG data of a single trial can be transformed into the following expression:

$$\mathbf{Z} = \mathbf{F}\mathbf{E} \quad (37)$$

Finally, f_p can be obtained from $\mathbf{Z}_p$ ($p = 1, \dots, 2m$) as the features of the original EEG data, expressed as shown in formula (38).

$$f_p = \log(\text{Var}(\mathbf{Z}_p) / \sum_{i=1}^{2m} \text{Var}(\mathbf{Z}_i)) \quad (38)$$

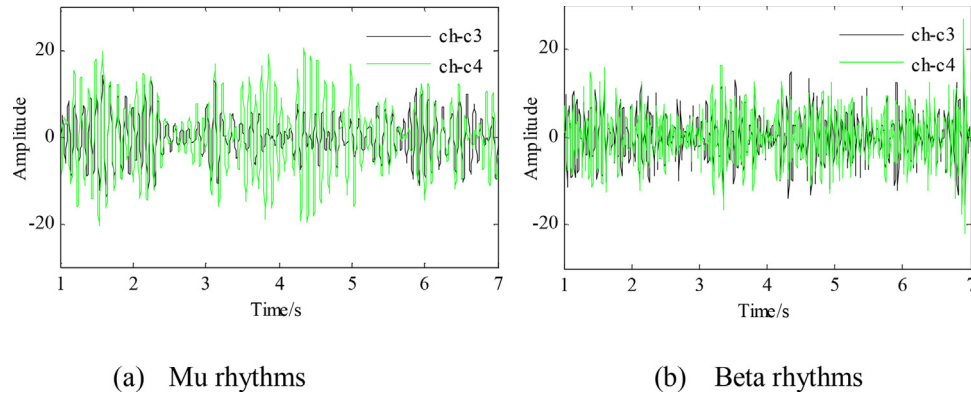


Fig. 3. Amplitude changes in beta and mu rhythms for right hand MI task of subject A08.

As the EEG signals of the BCI competition IV 2 B data set included only 3 channels, we set $m = 1$ for data set 2B. According to our experimental results, the best results can be achieved when $m = 2$ is used for data set 2A. Accordingly, in the following experiments, we used $m = 2$ for the 2A data set. The CSP features from different frequency bands were selected by the cross-validation results on training data sets. Fig. 4 is the visualized presentation of the CSP features from the 8–32 Hz frequency band for subject B04.

In Fig. 4, the blue symbol and green symbol represent the training and testing data respectively for motor imagery of the left hand. The purple and red symbols show the training and testing data respectively for motor imagery of the right hand. The data have separability, but were not linearly separable. Therefore, we employed several non-linear kernel functions to improve the classification performance.

4. Motor imagery pattern recognition results analysis

To evaluate the performance of the proposed classifier, we considered the classification accuracy and CPU running time as the index measure in the following experiments. All the experiments were implemented in Matlab 2014a on Windows 10, with an Intel Core i7 CPU (2.7 GHz) with 8-GB RAM. There were 3 experiments to evaluate the proposed MI pattern recognition methodology. The experimental implementations are described in the following sections.

4.1. Comparison of different kernel functions for the LSTSVM classifier

In this part of our work, we investigated linear, polynomial, sigmoid, wavelet and Gaussian kernel functions, as defined in formulas (18–25), for the proposed LSTSVM classifier, using the 2A and 2B data sets.

- In formula 20, we set c equal to 0.1, with d as the variable parameter for polynomial kernel.
- In formula 21, we set c equal to 0.1, with g is the variable parameter for sigmoid kernel.
- As defined in formula 22, the wavelet kernel is a new kernel based on wavelet analysis that had proven efficient in [47]. In formula 22, a is the variable parameter, and d is the dimension of an EEG sample.
- In the Gaussian radial kernel function, σ is the variable.

We used 10-fold cross validation for the classification results on the training data set to select the optimal variable parameters. Tables 1 and 2 show the classification results for data sets 2A and 2B

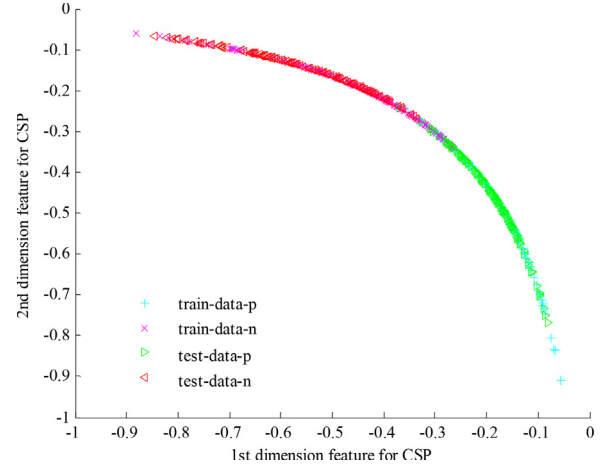


Fig. 4. 8–32 Hz CSP features for subject B04.

Table 1

Classification accuracy results using different kernel functions for data set 2A.

Subjects	Linear	Polynomial	Sigmoid	Wavelet	Gaussian
A01	84.03	75.69	72.22	72.22	88.19
A02	61.81	53.47	62.50	59.03	64.58
A03	89.58	87.50	90.28	88.19	94.44
A04	59.03	59.03	59.72	67.36	65.97
A05	65.97	65.07	64.58	70.14	76.39
A06	70.14	67.36	70.14	72.92	67.36
A07	70.83	71.53	74.31	72.22	75.00
A08	90.97	84.03	92.36	88.19	88.19
A09	90.97	86.11	79.86	75.00	88.89
Mean	75.93	72.20	73.99	73.92	78.78
Std	12.98	12.14	11.62	9.72	11.41

Table 2

Classification accuracy results using different kernel functions for data set 2B.

Subjects	Linear	Polynomial	Sigmoid	Wavelet	Gaussian
B01	60.00	60.31	62.50	60.31	65.63
B02	55.71	55.71	58.21	58.57	58.57
B03	54.38	52.50	52.50	55.31	59.38
B04	88.75	88.75	86.56	90.00	94.06
B05	58.44	55.94	69.38	73.44	73.44
B06	79.06	78.75	80.63	78.75	79.06
B07	75.31	75.00	75.00	71.88	81.25
B08	87.81	87.81	79.69	85.00	85.63
B09	76.25	75.94	76.25	70.94	76.88
Mean	70.63	70.08	71.19	71.58	74.88
Std	13.68	14.18	11.37	11.92	11.93

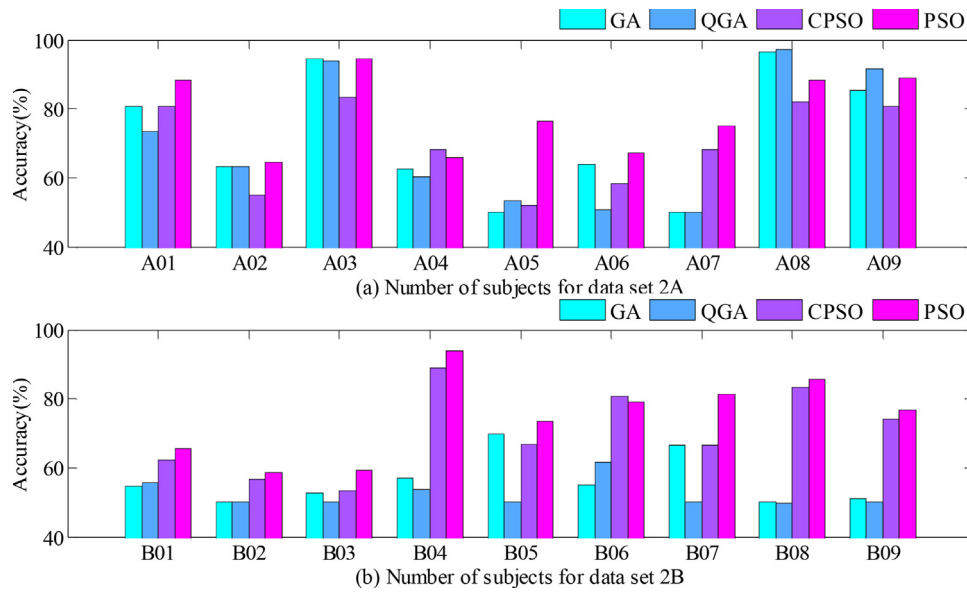


Fig. 5. Classification accuracy results using different optimization methods.

respectively. The optimal classification results varied for different subjects. As can be seen from Table 1, using Gaussian kernel function, the optimal accuracies for 5 out of 9 subjects were higher than for others. Furthermore, the average accuracy was 78.78%, which was higher than the others. For data set 2B, the linear, polynomial and Gaussian kernel functions achieved comparable performance insofar as 5 out of 9 experimental results were superior. Similar to the result for database 2A, the average accuracy was higher using Gaussian kernel function. In the following experiments, we used a Gaussian kernel for the LSTSVM classifier because it had greater generalization ability on the BCI Competition IV 2A and 2B data sets.

4.2. Parameter tuning for the LSTSVM classifier

There are many bio-inspired optimization algorithms and related improved techniques that have proven successful for SVM-based classifiers [14,45]. However, the performance of the convergence speed and the optimal classification results were affected by the classifiers and the data sets that we used. To achieve the best MI pattern recognition results, we investigated the PSO, CPSO, GA, and QGA algorithms for the LSTSVM classifier.

The proposed Gaussian kernel based LSTSVM classifier include 3 parameters: 2 penalty parameters c_1 and c_2 , and 1 Gaussian kernel parameter σ . For brevity's sake, we set $c_1 = c_2 = c$ for the classifier. The optimal values for penalty parameters c and kernel parameter σ were selected from the following range: $c \in \{10^{-8}, \dots, 10^4\}$, $\sigma \in \{2^{-4}, \dots, 2^8\}$.

In the 4 optimal algorithms, the maximum generation was set to 300, and 10-fold cross validation classification results of the training data set are the fitness or the evaluation criteria for selecting the optimal parameters. The population size was 40, the length of an individual was 20, and the generation gap was set to 0.95 for the GA-based model. The probabilities of crossover and mutation were set to 0.7 and 0.01, respectively. For parameter tuning of the classifier, the QGA combined the GA with quantum mechanics inspired from quantum physics. The implementation process included initialization the population, encoding parameters of the classifier, and calculating the fitness, selection, variation, evaluation and replacement. A quantum rotation gate strategy was used

for varying operators of the evolutionary algorithm. For comparison of the performance, the population size and the length of individuals for the QGA were same as for the GA. The quantum rotation angle in the QGA model was set to $0.01 \times \pi$.

A PSO algorithm was applied with the following parameters: swarm size equaled 20, and acceleration constants c_1 and c_2 were set to 1.5 and 1.7, respectively. In the particle initialization phase of the CPSO-based model, chaotic initialization was used to select a preferable initial position as compared to random selection. In the optimal position search process, the chaotic queues were generated based on the global optimal position searched by all particles. Then the particle position in the population was replaced by the best position determined from the chaotic queue. In this study, the CPSO algorithm was applied with the following parameters: swarm size n equaled 20, acceleration constants c_1 and c_2 were all equal to 1.5, and $r1$ and $r2$ were numbers generated by random distribution. The CPSO terminated after 300 iterations.

The testing results for data sets 2A and 2B are shown in Fig. 5 (a) and (b), respectively. The PSO, CPSO, and GA had comparable results, and they were better than QGA under the 300 iterations for data set 2A. However, using data set 2B, the PSO and CPSO achieved remarkable better results compared to the GA and QGA algorithms. Fig. 6 shows the average CPU running time of the iteration process. It is clear that PSO had the highest computational efficiency. Furthermore, PSO is simple, providing easier real-time hardware implementation. Therefore, we selected the PSO-based LSTSVM classifier for our proposed MI-BCI system.

4.3. Comparison of different TWSVM classifiers

As mentioned above, SVMs have outperformed most other classifiers in a wide variety of science and engineering applications, and their improved versions have proven to be successful in bioinformatics areas. In this section, we provide an explanation of the experiments we conducted to demonstrate the performance of our PSO-based LSTSVM classifier. We compared the LSTSVM classifier with SVM, TWSVM, and regularization projection twin SVM (RPTSVM) approaches, and we now provide details regarding all of the experiment settings. The penalty parameter c was selected from $10^{-8} - 10^4$. For simplicity, we set $c_1 = c_2 = c$ for all the TWSVMs,

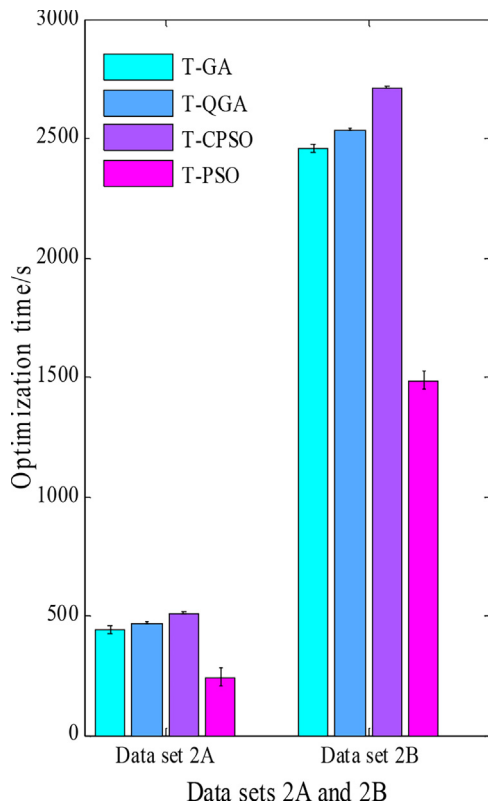


Fig. 6. Comparison of the optimization times.

$v_1 = v_2 = v$ for the RPTSVMs, and the bounded parameters v in the RPTSVMs were taken from the set $\{0.1, \dots, 0.9\}$.

The optimal parameters, classification accuracy, and CPU running times are summarized in Tables 3 and 4 on the two data sets BCI competition IV 2A and 2B. The classification accuracy (Acc) was obtained by training the classifiers on the training data sets, with an evaluation on the test data sets. The optimal accuracy (opti-acc) for the training data set is also shown in the tables, but it represents only the cross validation optimal results of the training data. The best accuracy and fastest running time for each subject are shown in bold.

From Tables 3 and 4, it is clear that the training times of the LSTSVM were highly competitive with the other SVMs, and it is superior to the other supervised machine learning methods in terms of hardware online implementation. Using the LSTSVM classifier, 7 of the 9 total subjects demonstrated an improvement as shown in Table 3. In Table 4, the results are that 4 of the 9 total subjects are superior. For data set 2A, the average classification accuracy of our proposed method showed an increase of 3.87%, 6.71%, and 6.10% over the results from PSO-SVM, PSO-TWSVM, and PSO-RPTSVM, respectively. Our proposed classifier also achieved a faster average training time of 0.0020s. As can be seen from Table 4, the mean classification accuracy of the proposed method was greater by 4.45%, 5.46%, and 4.73% compared to PSO-SVM, PSO-TWSVM, and PSO-RPTSVM, respectively. Using the data set 2B, we also achieved fast training and testing times of 0.0218 s and 0.0242s.

4.4. Comparison of popular motor imagery task classification methods

In this subsection, we compare the PSO-LSTSVM classifier with the published machine learning classifiers used for MI task recognition, such as linear discriminant analysis (LDA), back propagation (BP) and probabilistic neural network (PNN). In our experiments,

the layers and parameters for the BP and PNN were regulated to be the best fit in the classification task. For the BP classifier, the learning rate was 0.1, and its ratio was 0.01 after 5 loops. The number of hidden layer nodes of the 3-layer BP network was selected by 5-fold cross validation on training data. The spreading coefficient in the pattern layer of the PNN classifier was tuned by 5-fold cross validation on training data.

The experimental results are summarized in Table 5, where the best results among the 4 different methods are indicated in bold. Using data set 2A, the mean accuracy of PSO-LSTSVM, LDA, BP, and PNN was 78.78%, 74.77%, 76.23%, and 73.92%, respectively. The average accuracy of our proposed method showed an improvement of 4.01%, 2.55%, and 4.86% over the other three classifiers. Similarly, the average classification accuracy of the proposed method on data set 2B also showed an increase of 4.10%, 8.62%, and 4.27%, respectively. On average, our proposed method significantly outperformed the competing methods. Clearly, the overall performance of PSO-LSTSVM and LDA was competitive with BP and PNN.

5. Discussion

The artifact removal, feature extraction and pattern classification play a significant role for an MI-BCI system. Self-adaptive de-noising and self-adaptive frequency selection for CSP feature extraction were used to improve the classification performance as well as decrease the inter-subject variations. We also introduced the least squares version of the TWSVM to the MI-BCI system for the first time. Furthermore, we used PSO, CPSO, GA, and QGA to select the optimal parameters for the classifier.

We investigated the recently published MI pattern recognition literature on BCI competition IV data sets 2A and 2B. In one study [17], the authors used a covariate shift-detection method for motor imagery-based BCI. They achieved a mean accuracy of 74.07%, 74.92%, and 73.84% for their TLCSD₁, TLCSD₂, and ALCSO methods, respectively, on dataset 2A and achieved 66.15%, 69.72%, and 67.88% accuracy for their TLCSD₁, TLCSD₂ and ALCSO methods, respectively, on data set 2B. As is shown in Table 5, our experimental results achieved a mean accuracy of 78.78% and 74.88% on data sets 2A and 2B, respectively.

In another study [21], a novel Bayesian framework for discriminative feature extraction and classification was performed on data set 2A. Compared with their experimental results under the same CSP-based preprocessing method, the mean classification accuracy of our proposed method showed an increase of 5.32%, and 6 out of 9 of our classification results were superior to their reported experimental results.

H.W.Mo et al. [14] used a magnetic bacteria optimization-based SVM for motor imagery EEG classification on the BCI competition IV 2A data set for 4 subjects. However, they performed the 4 classes of left hand, right hand, foot and tongue motor imagery tasks. The CPU running time of the MI pattern recognition system had not been reported in the above publications.

In summary, our proposed method yielded higher overall classification accuracy and decreased inter-subject variations. However, the optimal accuracy results on the training data and the final testing accuracy results on the evaluation data were quite variable, as shown in Tables 3 and 4. These results demonstrated that the subjects exhibited substantial inconsistency in performance over the course of individual sessions. The classification performance for each individual subject varied greatly across session times. Recent studies regarded neural processes as the potential causes of sensorimotor rhythms. The variations in activity between different attentional networks lead to the variations performance of the SMR-BCI.

Table 3
Classification performance from BCI competition IV data set 2A.

Subjects	Methods	Optimal parameters			Opti-acc(%)	t.train (s)	t.test (s)	Acc (%)
		C	σ	ν				
A01	PSO-SVM	0.8040	6.7343		87.02	0.0523	0.0183	78.32
	PSO-TWSVM	1×10^{-8}	14.6626		88.19	0.0184	0.0121	83.33
	PSO-RPTSVM	1×10^{-8}	189.2577	0.9000	86.33	0.0238	0.0121	80.56
	PSO-LSTSVM	1.0537	0.0625		93.80	0.0019	0.0098	88.19
A02	PSO-SVM	1.0000	71.1904		80.92	0.0393	0.0246	63.89
	PSO-TWSVM	1×10^{-8}	0.0625		77.00	0.0299	0.0102	52.08
	PSO-RPTSVM	8.5085×10^3	256	0.1000	75.68	0.0287	0.0108	56.94
	PSO-LSTSVM	1×10^{-8}	256		81.38	0.0019	0.0104	64.58
A03	PSO-SVM	0.0823	123.2085		97.50	0.0585	0.0162	86.81
	PSO-TWSVM	10000	9.4022		98.71	0.0325	0.0102	91.67
	PSO-RPTSVM	1×10^{-8}	256	0.1000	98.04	0.0225	0.0137	91.67
	PSO-LSTSVM	1×10^{-8}	24.1217		98.00	0.0023	0.0101	94.44
A04	PSO-SVM	0.7615	1.7543		88.19	0.0415	0.0162	64.58
	PSO-TWSVM	1×10^{-8}	0.0625		86.05	0.0270	0.0102	65.97
	PSO-RPTSVM	1×10^{-8}	0.0625	0.5473	87.55	0.0229	0.0089	63.19
	PSO-LSTSVM	1×10^{-8}	229.1975		84.05	0.0021	0.0091	65.97
A05	PSO-SVM	0.0919	88.7742		80.81	0.0590	0.0178	67.36
	PSO-TWSVM	1×10^{-8}	0.0625		79.29	0.0265	0.0087	56.25
	PSO-RPTSVM	1×10^{-8}	0.0625	0.9000	80.69	0.0243	0.0085	59.03
	PSO-LSTSVM	5.5956×10^3	5.1462		94.24	0.0019	0.0108	76.39
A06	PSO-SVM	0.6882	6.4559		85.52	0.0453	0.0220	63.19
	PSO-TWSVM	1×10^{-8}	0.0625		82.55	0.0267	0.0090	66.67
	PSO-RPTSVM	9.4623×10^3	256	0.4140	84.70	0.0273	0.0089	63.19
	PSO-LSTSVM	1×10^{-8}	0.0625		85.24	0.0020	0.0103	67.36
A07	PSO-SVM	0.9545	3.9789		89.10	0.0742	0.0150	72.22
	PSO-TWSVM	1×10^{-8}	0.0625		86.90	0.0258	0.0094	62.50
	PSO-RPTSVM	4.9460×10^3	61.1022	0.3600	86.81	0.0249	0.0089	64.58
	PSO-LSTSVM	1×10^{-8}	256		81.98	0.0018	0.0101	75.00
A08	PSO-SVM	1.0000	2.0516		98.66	0.2298	0.0237	91.67
	PSO-TWSVM	10000	1.2405		99.38	0.0395	0.0088	87.50
	PSO-RPTSVM	10000	0.7826	0.1000	98.75	0.0475	0.0105	92.36
	PSO-LSTSVM	1×10^{-8}	201.4625		99.38	0.0022	0.0090	88.19
A09	PSO-SVM	1.0000	7.5949		85.64	0.0526	0.0159	86.11
	PSO-TWSVM	1×10^{-8}	0.9148		86.13	0.0242	0.0086	82.64
	PSO-RPTSVM	1×10^{-8}	0.3785	0.9000	84.26	0.0377	0.0099	82.64
	PSO-LSTSVM	1×10^{-8}	101.0968		78.52	0.0018	0.0099	88.89
Mean	PSO-SVM	*	*		88.15	0.0725	0.0189	74.91
	PSO-TWSVM	*	*		87.13	0.0278	0.0099	72.07
	PSO-RPTSVM	*	*	*	86.98	0.0288	0.0102	72.68
	PSO-LSTSVM	*	*		88.51	0.0020	0.0099	78.78
Std	PSO-SVM	*	*		6.32	*	*	11.12
	PSO-TWSVM	*	*		7.69	*	*	14.44
	PSO-RPTSVM	*	*	*	7.43	*	*	14.09
	PSO-LSTSVM	*	*		7.85	*	*	11.41

6. Conclusion

Machine learning has considered a useful tool for the development of BCI research. While machine learning continues to make important contributions to the field, three key problems continue to present challenges for classifying sensorimotor-rhythms, thereby preventing the real-world application of BCIs. First, the frequency bands of ERD/ERS patterns are variable across subjects and across event trials for the same subject. Second, EEG signals tend to be contaminated with artifacts and noise that can cause performance degradation in classification. Third, self-adaptive training strategies for the classifier needed to adapt BCIs to different users still need enhancement.

In this study, we proposed a novel MI-BCI system that used a self-adaptive frequency selection CSP combined with PSO LSTSVM classifier scheme. The adaptive artifact removal and inter-subject frequency band selection CSP were employed for EEG signal preprocessing and feature extraction. To evaluate the proposed LSTSVM classifier for MI patterns, we investigated linear, polynomial, sigmoid, wavelet, and Gaussian kernel functions. Our experimental results showed that the Gaussian kernel had more generalization ability on the BCI competition IV data sets 2A and 2B. To achieve

the optimal recognition results, we also compared several biology-inspired algorithms and their improved versions for MI pattern recognition using the same public data sets.

The experimental results demonstrated that our proposed method increased training efficiency, and classification accuracies was increased as well. Furthermore, the standard deviation of inter-subject accuracy decreased compared with other competing methods. Therefore, we conclude that our proposed MI pattern recognition scheme is suitable for the on-line self-adaptive study of BCI systems. According to these results, the training strategies need to be further enhanced to improve the classification performance and decrease the within-subject variations. Future work will include investigation of the neuro-physiological causes of such variations, as well as exploration of the best on-line training mechanisms for adjusting the self-adaptive classifier to accommodate different types of subjects.

Ethical statements

We validated our method by an open-access BCI dataset, i.e. the dataset IIa and IIb from BCI competition IV (<http://www.bbci.de/competition/iv/>). Thus, our study did not involve any self-recorded

Table 4

Classification performance from BCI competition IV data set 2B.

Subjects	Methods	Optimal parameters			Opti-acc(%)	t.train (s)	t.test (s)	Acc (%)
		c	σ	ν				
B01	PSO-SVM	0.0182	209.7649		72.50	0.2321	0.0567	57.19
	PSO-TWSVM	1×10^{-8}	18.7940		72.50	0.1005	0.0220	59.06
	PSO-RPTSVM	1×10^{-8}	0.0625	0.9000	72.25	0.0825	0.0196	62.19
	PSO-LSTSVM	1×10^{-8}	159.1530		64.25	0.0176	0.0230	65.63
B02	PSO-SVM	1.1358×10^{-4}	51.0225		61.00	0.1276	0.0342	58.21
	PSO-TWSVM	1×10^{-8}	0.2159		62.00	0.0689	0.0199	58.57
	PSO-RPTSVM	1×10^{-8}	0.0625	0.1000	62.00	0.0717	0.0168	50.71
	PSO-LSTSVM	1×10^{-8}	244.4722		63.26	0.0216	0.0263	58.57
B03	PSO-SVM	20.3943	9.1686		59.00	0.0860	0.0460	52.50
	PSO-TWSVM	1×10^{-8}	5.0025		59.00	0.0690	0.0199	59.69
	PSO-RPTSVM	1×10^{-8}	220.3445	0.1000	58.25	0.0675	0.0177	54.38
	PSO-LSTSVM	1×10^{-8}	99.9196		58.50	0.0331	0.0247	59.38
B04	PSO-SVM	0.8924	153.6625		87.33	0.1077	0.0255	89.69
	PSO-TWSVM	1×10^{-8}	0.5639		90.71	0.0748	0.0211	95.31
	PSO-RPTSVM	1×10^{-8}	0.5639	0.5264	91.90	0.0991	0.0180	86.88
	PSO-LSTSVM	6.3206×10^3	16.5000		90.95	0.0208	0.0234	94.06
B05	PSO-SVM	0.1000	16.0000		70.00	0.1205	0.0516	69.38
	PSO-TWSVM	1×10^{-8}	13.4452		70.00	0.0877	0.0250	51.52
	PSO-RPTSVM	152.0215	253.3820	0.1014	70.24	0.1110	0.0196	70.00
	PSO-LSTSVM	1×10^{-8}	256.0000		74.76	0.0213	0.0242	73.44
B06	PSO-SVM	0.1000	0.2500		74.00	0.1839	0.0505	79.69
	PSO-TWSVM	1×10^{-8}	0.0625		76.25	0.0866	0.0213	76.25
	PSO-RPTSVM	1×10^{-8}	256	0.1000	75.25	0.0766	0.0188	78.75
	PSO-LSTSVM	1×10^{-8}	144.5748		76.00	0.0170	0.0222	79.06
B07	PSO-SVM	0.1000	0.2500		68.00	0.1443	0.0270	65.63
	PSO-TWSVM	1×10^{-8}	0.3935		69.50	0.0729	0.0241	65.31
	PSO-RPTSVM	7.2632×10^3	154.2108	0.9000	67.75	0.0945	0.0223	69.69
	PSO-LSTSVM	1×10^{-8}	112.1374		72.25	0.0168	0.0219	81.25
B08	PSO-SVM	0.1000	12.4849		68.64	0.1633	0.0526	87.50
	PSO-TWSVM	1×10^{-8}	1.0279		70.68	0.0906	0.0214	82.81
	PSO-RPTSVM	1×10^{-8}	0.0625	0.4876	70.00	0.0948	0.0196	82.50
	PSO-LSTSVM	1×10^{-8}	0.0625		71.82	0.0249	0.0257	85.63
B09	PSO-SVM	1.9154	0.2500		70.75	0.2048	0.0213	74.06
	PSO-TWSVM	1×10^{-8}	4.1119		70.00	0.0922	0.0205	76.25
	PSO-RPTSVM	1×10^{-8}	2.9938	0.9000	69.75	0.0817	0.0185	76.25
	PSO-LSTSVM	7.9140×10^3	9.2811		70.75	0.0230	0.0287	76.88
Mean	PSO-SVM	*	*		70.14	0.1522	0.0406	70.43
	PSO-TWSVM	*	*		71.18	0.0826	0.0220	69.42
	PSO-RPTSVM	*	*	*	70.82	0.0866	0.0190	70.15
	PSO-LSTSVM	*	*		71.35	0.0218	0.0242	74.88
Std	PSO-SVM	*	*		8.15	*	*	13.38
	PSO-TWSVM	*	*		9.00	*	*	14.14
	PSO-RPTSVM	*	*	*	9.44	*	*	12.42
	PSO-LSTSVM	*	*		9.39	*	*	11.93

Table 5

Classification performance of different machine learning methods.

Subjects (ComIV 2A)	Methods				Subjects (ComIV 2B)	Methods			
	ProposedAcc(%)	LDAAcc(%)	BPAcc(%)	PNNAcc(%)		ProposedAcc(%)	LDAAcc(%)	BPAcc(%)	PNNAcc(%)
A01	88.19	88.89	72.92	84.03	B01	65.63	58.13	68.13	57.81
A02	64.58	58.33	62.50	61.11	B02	58.57	56.07	50.36	56.43
A03	94.44	91.67	98.61	88.89	B03	59.38	52.81	50.63	54.38
A04	65.97	63.89	71.53	70.14	B04	94.06	89.06	93.44	89.69
A05	76.39	54.86	52.08	50.00	B05	73.44	69.38	53.75	68.75
A06	67.36	70.14	75.00	65.28	B06	79.06	78.75	59.38	78.44
A07	75.00	70.83	74.31	58.33	B07	81.25	69.38	63.44	65.63
A08	88.19	90.28	90.97	95.83	B08	85.63	87.50	86.88	88.13
A09	88.89	84.03	88.19	91.67	B09	76.88	75.94	70.31	76.25
Mean	78.78	74.77	76.23	73.92	Mean	74.88	70.78	66.26	70.61
Std	11.41	14.29	14.45	16.54	Std	11.93	13.26	15.38	13.33

datasets from human participants or animals. The authors declare that they have no conflict of interest.

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